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Reliability-Driven Cloudlet Allocation Framework Leveraging Chaotic Evolutionary Techniques for Mobile Cloud Environments

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ABSTRACT: Cloud computing provides cost-efficient opportunities for enterprises by offering a variety of dynamic, scalable, and shared services. Usually, cloud providers provide assurances by specifying technical and functional descriptions in Service Level Agreements (SLAs) for the services they offer. The descriptions in SLAs are not consistent among the cloud providers, even though they offer services with similar functionality. Therefore, customers are not sure whether they can identify a trustworthy cloud provider only based on its SLA. Propose a Trust-Based Cloudlet Service Selection (TCSS) system architecture for Mobile Cloud Computing (MCC), a marketplace. Also, Chaos Genetic Algorithm (CGA) is proposed for optimizing the fitness value for trusted selection of cloud services by taking into account the mobile user's trust expectation. This system provides means to identify the trustworthy cloud providers in terms of different attributes (e.g., security and performance) assessed by multiple sources and roots of trust information.

KEYWORDS: Mobile Cloud Computing (MCC), Chaos Genetic Algorithm (CGA), Trust-Based Cloudlet Service Selection (TCSS), Cloudlet and Service Level Agreements (SLAs).

1 Introduction

Mobile cloud computing (MCC) is a new computing model where cloud computing (CC) is integrated into mobile computing environments. MCC is derived from cloud computing; thus, it inherits the advantages of cloud computing, such as dynamic development of mobile applications, resource scalability, multiuser sharing, and multiservice integration. The new computing model breaks through the resource limitations of mobile terminals by moving data processing and storage from mobile devices to cloud service platforms via wireless networks. Rich mobile applications can be easily created and accessed just based on a web browser on mobile devices [1].

In MCC, computation-intensive tasks can be migrated to the resource-rich cloud nodes so as to improve application performance and reduce energy consumption of mobile devices. In order to further reduce the cloud access delay, infrastructure-based cloudlet has been introduced, which acts as the middle layer between the cloud and mobile users. Since it is deployed near Wi-Fi access points, it can provide efficient cloud resources for nearby mobile users with lower latency. A cloudlet is located at the edge of the Internet is a mobility-enhanced small-scale cloud data center, which extends cloud computing infrastructure [2]. But, quite apart from that, it also has some problems, including resource limits, weak battery strength, user mobility, and low network coverage. However,



with the rapidly growing density of mobile devices, such a kind of cloudlet will also face the resource limitation challenge.

In the existence of the MCC, a large number of users of the internet through their digital assistant make use of the cloud technology via the wireless interface. The selection of the prompt cloud service becomes more trivial and an important issue to be addressed. Personal digital assistants are used for many online services that distribute a range of files in the cloud. These get access to the cloud environment using any of the technologies, such as 4G, LTE, GPRS, that give rise to different challenges, such as bandwidth limitation, cost, capacity to store, and latency.

In this study, a method that is efficient and more secure mobile cloud computing model based on the Cloudlet concept is presented, the mobile device users can connect directly to cloud resources. The proposed cloudlet-based model is developed in such a way that it can be used across a range of applications where there is a high need for efficiency and security. In which the initial population is generated by chaos mapping and refined by competition within each iteration of this approach, in addition to the evolution of the genetic algorithm (GA) [4].

2 Literature Review

Naseer et al [5] introduced a model that can help the users of the cloud services (CSUs) in finding a service provider (CSPs) that is more efficient and trusted enough based on the data that is gathered from the statutory authorities, CSP's performances, and more from the feedback given by the consumers. The model was also customizable as per the changing requirements of the end users, based on the level of precedence of the users. The demands that are mostly preferred by the users are taken into consideration while developing such a trustworthy model in the service layer of the cloud technology.

Manuel [6] proposed a unique technique that is based on the credentials recorded in the past and the current abilities of a service provider. The four parameters, namely the availability, reliability, turnaround efficiency, and data integrity, were considered as the key for the calculation of the trust value. It depicted how an SLA (Service Level Agreement) is made between the user and the provider in terms of the requirements of the user and the abilities of a cloud service provider. The results proved that this model outperforms the [5] and other competitive models.

Saravanan and Vani [7] elucidated a framework shortly termed as SelCSP that makes a combination of trust and the level of competence for estimating the risk of interacting for supporting the consumers in identifying the best service provider. The measure of trust is calculated from the personal experiences that are gained by the one-one interactions or through the feedback that is related to the reputation of a service provider. Transparency is the factor that determines the competence of an SLA. A case study was also made to present the proposed approach.

Emeakaroha [8] presented the operationalization of a trust label system designed to communicate trust and trustworthiness in Cloud services to improve consumer confidence and trust in Cloud providers. It describes the technical details and implementation of the trust label components. Based on a use case scenario, an initial evaluation was carried out to test its operations and its usefulness for increasing consumer trust in Cloud services.

Billhardt [9] investigated the problem of vendor selection with the use of trust and the techniques based on reputation. Many of them are based on the consumers' direct experiences they had with the application delivered through a vendor. When there is no experience for reference, the acquaintances of the consumer are contacted, as they serve as an input for the reputation of a service provider. On the contrary, the work urges for an approach that is based on the experience for the selection of vendors where the consumers place trust and mechanisms of reputation for the future service providers from a situation that is similar.

Agheli et al [10] came up with a unique issue from where it was clear that the major obstacle in bringing the cloud technology into a wide range of use is the lack of consumers' trust in the vendors. The trust creation issue is addressed in this research, and the outcome was a predictive model based on the priority of the services that are promised by the vendor and allowing the users to validate and choose a suitable platform. Based on the trust model of subjective logics, the research made a combination of the experiences, reputation values, satisfied interaction feedback, and risk factors.

Sharma and Mahrishi [11] put forth a trust-based model that depends on the authorizations that were made in the past and on the current competency of a vendor. The value for the trust is based on the capability, feedback, and behavior. The results indicate that the proposed model is found to be better than the other existing models for building trust in the consumers.

Khurana and Bawa [12] presented a Quality-Based Cloud Service Broker framework (QCSB) will be proposed. This framework will not only help Cloud consumers (CC) to select optimal Cloud service providers (CSP) but also rank candidate CSPs according to user preferences mentioned in SLA. Subsequent sections will discuss its components, algorithm, and implementation issues. A case study on three CSPs with three Quality metrics will be executed using the CloudSim simulator. It is envisioned that the present research will benefit all those users who are interested in SLA based selection and ranking of CSPs.

Mangla et al [13] proposed a trust management framework for effective selection of Cloud providers to fulfil numerous Cloud users' requirements reliably. Due to a lack of trust between these two stakeholders has delayed the worldwide acceptance of Cloud for outsourced facilities has been delayed. The TM framework for effective matching of Cloud facilities to fulfill numerous Cloud users' requirements is presented. Also, an outline is made to compute the trustworthiness of any service provider based on the individual perception, collaborative decision, and on the credential characteristics of the provider itself.

Li and Ping [14] investigated the various models of trust that are used in a very large and distributed atmosphere and further introduced a unique model for the trust factor for addressing the issues in the security of a cloud service, where the consumer is independent in choosing the service provider that is heterogeneous and compatible. The proposed model is based on the domain. Here, an agent is set for building trust by dividing the resource nodes within the cloud. In this model, the recommendations are considered as one of the services in the cloud, similar to a computational service or a storage service. The experiments prove that it can build an efficient trustworthiness model in a cross-cloud atmosphere.

3 Proposed Methodology

In general, MCC is growing rapidly, with new providers will increasingly compete for customers by providing services with trust and performance. In this work, we introduced a chaos genetic algorithm [15] for optimizing the trust value for selecting a cloud with trustworthy and good performance. The selected cloud services for chromosome to calculate the trust value are (i) cloud certified by a standard agency, and (ii) cloudlet in the same region considered for trust ranking, and (i) power consumption and (ii) response time will be considered for performance ranking. The total ranking of service selection of the service provider is calculated by the sum of trust ranking and performance ranking. Any requests for the service by a user are forwarded by the mobile network to the server in the cloud post the process of validation, confirming the legitimacy of the user. The server then processes the request one at a time. The cloudlet scheme is widely used to resolve many issues related to the MCC. The cloudlet is a PC that is equipped with the required specifications and is made accessible by PDAs in a given area/range. The overall layout of the proposed method is depicted in Figure 1. Here, the services are offered from a remote place to the users who are in close distance in

a minimum time with a great throughput. The environment with three cloudlets is shown in Figure 2.

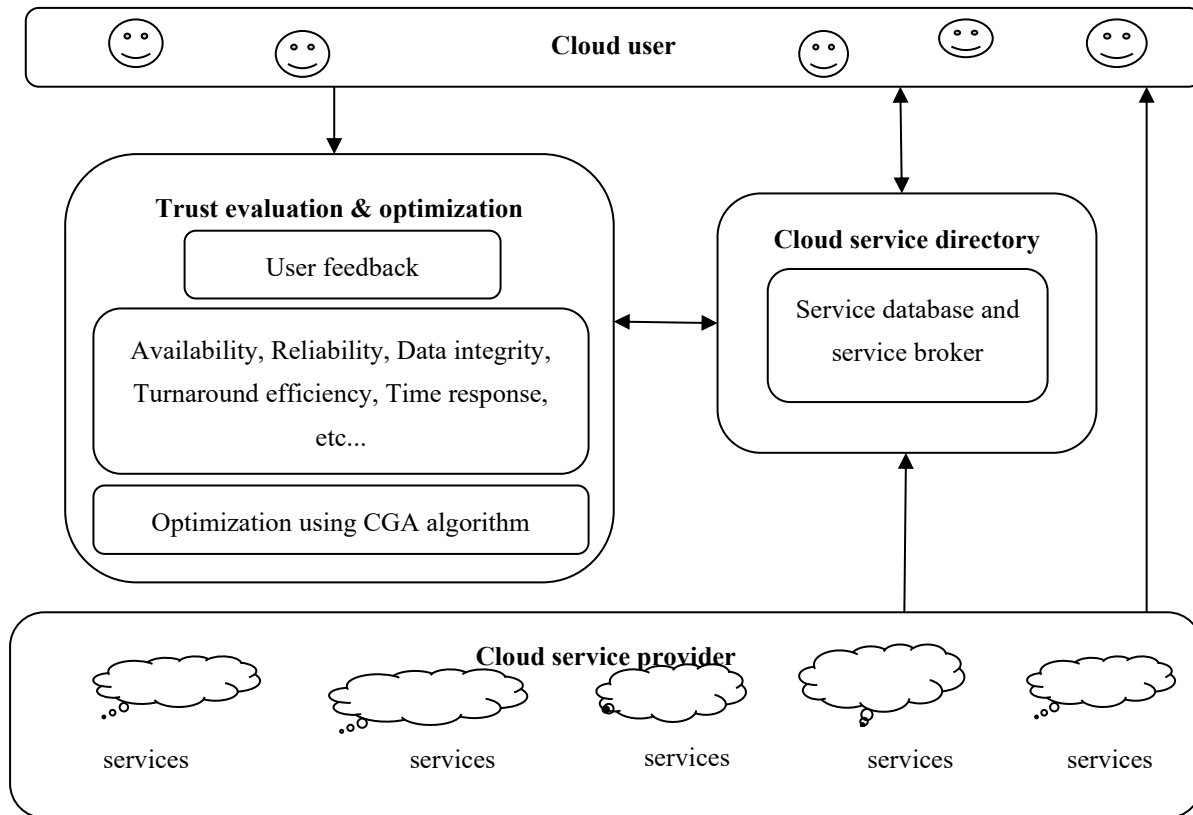


Figure 1: Overview of proposed methodology

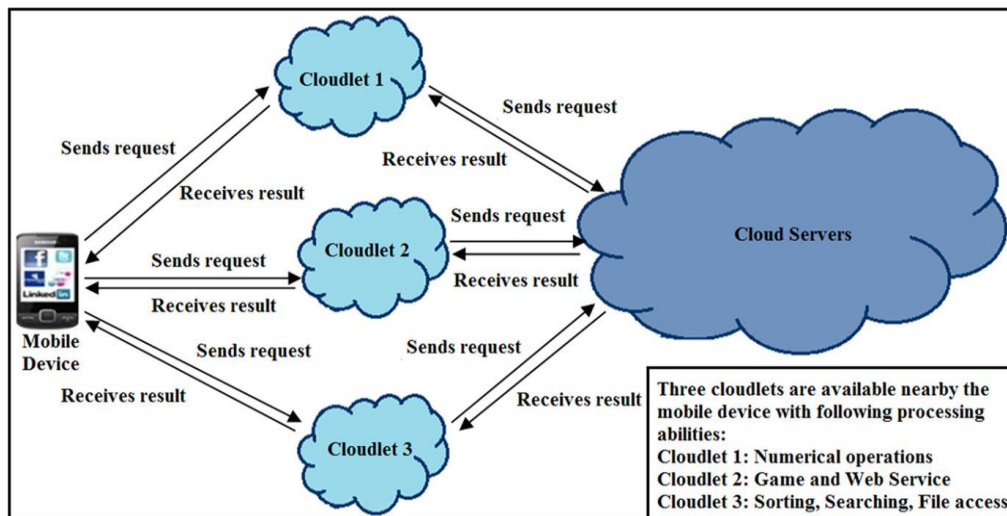


Figure 2: Multi-cloudlet environments near the mobile device

It can be divided into three parts are layer 1, layer 2, and layer 3. Layer 1 consists of mobile devices, layer 2 consists of agent cloudlet, and layer 3 consists of cloud providing Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). Layer 1 consists of the mobile devices that place the requests for a service to layer 2, which has the cloudlet. The layer 2

provides the requested service in return if there is a possibility, else the requests are handed over to the actual cloud is present in the layer 3, which gets the requests and sends back the appropriate results to the cloudlet, and the results are again transferred to the PDAs in the layer 1. These are the common processes that take place in mobile cloud computing generally. The contribution of this work will take place between layer 2 and layer 3, based on trust for the selection of a cloudlet is called the trust-based service selection model. So, the cloudlet selection based on trust and performance will be discussed in the section below. As discussed earlier, trust will be measured by a cloud certified by a standard agency and a cloudlet in the same region, and the Performance value will be measured in terms of time response, and power consumption will be explained.

3.1 Service Selection based on Trust value

Here, the trust service provider will be calculated based on the user's feedback and service provider credentials. Trust Management Service (TMS): In a classic interaction of the TMS that is based on the reputation, feedback is given by the end user, or in another case, the user requests the assessment of the cloud service in view of the trust.

3.2 Feedback Collusion Detection

There are high chances for a great number of false feedbacks performed by a malicious user to manipulate the results of the cloud's trustworthiness. (feedback that is promoting and attacks of Slandering). It is suggested by a decent count of researchers to validate the feedback based on its count of positivity, but this alone cannot be considered for the determination of the credibility. Either feedback or an assessment of the trustworthiness is given by the end user. The behavior in terms of the trust of a service offered in the cloud is the collection comprised of the history of invocation, the feedback, and this forms the basics of various parameters of the Quality of Service (QoS). This happens in-between the service providers and the consumers. The CSPs are contacted, and information on the cost, the time complexity, and the type of service is stored; in addition, the users get connected with the brokers for the fulfillment of the requirements.

3.2.1 Cloud certified by a standard agency

The important cloud certifications by the standard agency of service providers are availability, reliability, data integrity, and turnaround efficiency, as given below [1]:

Availability:

Availability (AVI) is the degree to which the required services are accessible. Let $SP_{REQUEST}$ and $SP_{ACCEPTED}$ represent the number of service requests submitted to and accepted by the Service Provider over a period of time T. So, the availability can be determined as:

$$AV_{SP} = SP_{ACCEPTED} / SP_{REQUEST} \quad (1)$$

where $0 \leq AV_{SP} \leq 1$. As the value of AV_{SP} approaches 1, availability increases.

Reliability:

Reliability (RE) of a service provider is measured in terms of the successful completion of service requests. Let $SERV_{DELIVERY}$ denote the number of successful service deliveries by service provider over a period of time T, then the reliability of service PROVIDER can be determined as:

$$RI_{SP} = SERV_{DELIVERY} / SP_{ACCEPTED} \quad (2)$$

As the value of RI_{SP} approaches 1, reliability increases.

Data integrity

It refers to the security and privacy of data. Let α_{SP} denotes the requests successfully served with desired data security by the service provider over a period of time T, then the data integrity of the service provider can be determined by using.

$$DI_{SP} = \alpha / SERV_{DELIVER} \quad (3)$$

Turnaround Efficiency

Turnaround time is the time elapsed between the submission of the service request to the successful service delivery. It is an important parameter incorporated in the SLA between the service provider and the Cloud user. Turnaround Efficiency (TE) of the service provider can be evaluated as:

$$TE_{SP} = (\text{Promised turnaround time for a service in SLA}) / (\text{Actual turnaround time for the service}) \quad (4)$$

Let w_1, w_2, w_3 , and w_4 be the weights associated with AV, RE, DI, and TE, respectively, based on the priority of these attributes. The standard agency-based trust of service provider SA(SP) can be evaluated as:

$$SA^{SP} = w_1 * AV^{SP} + w_2 * RI^{SP} + w_3 * DI^{SP} + w_4 * TE^{SP} \quad (5)$$

Where $w_1 + w_2 + w_3 + w_4 = 1$

Service selection based on performance analysis consists of delay time and power consumption range.

The Trust Value (TV) of service provider 'SP' can be evaluated as:

$$TV = \mu_1 * DT^{SP} + \mu_2 * RT^{SP} + \mu_3 * SA^{SP} \quad (6)$$

Where $\mu_1 + \mu_2 + \mu_3 = 1$, and these weights are assigned based on the priority.

Direct Trust: Direct trust (DT^{SP}) between Cloud user (Cl_u) and Cloud Service provider (SP) is computed based on direct interactions.

$$DT_u^{SP} = \begin{cases} \frac{SP_{REQUEST_u}}{SP_{REQUEST_u} + \beta SP_{ACCEPTED_u}} * Td_u(t), & SP_{REQUEST_u} \geq SP_{ACCEPTED_u} \\ 0 & \text{else} \end{cases} \quad (7)$$

$SP_{REQUEST_u}$ represents the number of successful service delivery cases by the service provider and $SP_{ACCEPTED_u}$ corresponds to failure due to wrong credentials of the provider or non-attainment of specified Quality of Service (QoS) parameters. $\beta \geq 1$ is the punishment factor forcing the service provider to advertise authentic information. The weightage of user feedback is related to time; the longer the time is the less reliable the information available. Let $Td_u(t)$ is a time delay function representing the weightage of the u^{th} Cloud user at time 't' then:

$$Td_u(t) = \frac{1}{1 + \theta(t - t_i)} \quad (8)$$

where θ is the time delay speed factor, and t_i represents the time when the i^{th} feedback for the same provider was given. The cloudlet selection delay ' θ ' considers the following constraints

- (i) The time delay for the sending of messages to the cloudlets that are nearby (T_{send}),
- (ii) The time delay for receiving the responses from the cloudlets that are nearby (T_{recv}),

(iii) The time delay for accessing the information of the cloudlet from the table (CIT) (T_{access}).

The cloudlet selection delay is given as:

$$\theta = (T_{\text{send}}, T_{\text{recv}}, T_{\text{access}}) = T_{\text{send}} + T_{\text{recv}} + T_{\text{access}} \quad (9)$$

Recommended Trust: Recommended trust (RT^{SP}) for a service provider refers to the trust information obtained from other Cloud users who have interacted with the same service provider. It can be obtained by taking the average of all Cloud users having experience with the services of service provider 'SP'.

$$RT^{\text{SP}} = \frac{\sum_{i=1}^n DT_u^{\text{SP}}}{n} \quad (10)$$

These are the trust values evaluated for service selection in cloudlet, and to optimize of trust value, chaos genetic algorithm is used.

3.3 Genetic Algorithm (GA)

GA is widely used in many approaches for the representation of the process of diagnosis. These are used for representing and calculating the appropriate fitness for the optimization of the values of the trust for selecting a good vendor.

Genetic Algorithm [16] is a procedure to search that is available globally for the searching mechanism from a particular population to another. It is based on probability and is mostly applied to problems that are difficult for optimization and learning. These are the ones that can be seen in the natural evolution. This is used as an approach for searching globally for the best optimization. When the GAs is normally applied to the problem of optimizing, and when the requirement is found to be unsatisfactory, there forms a subset is formed that is appropriate. It then requires that the alternatives are to be coded in the form of strings of a definite length that have symbols and alphabets. These are termed chromosomes, and the symbols that arise out of this are termed genes, and the collection is termed a gene pool. The GAs generally searches for a suitable alternative in the sense of a particular fitness function over the evolution of the chromosomes. The preliminary steps involved in a GA are depicted in Figure 3.

GA is based on the biological concept of generating the population. According to Darwin's theory, the term "Survival of the Fittest" is used as the method of scheduling in which the tasks are assigned to resources according to the value of the fitness function for each parameter of the task scheduling process. The main principles of the GA are described as follows:

3.3.1 Initial Population

The initial population is the set of all individuals that are used in the GA to find the optimal solution. Every solution in the population is called an individual. Every individual is represented as a chromosome to make it suitable for the genetic operations. From the initial population, the individuals are selected, and some operations are applied to them to form the next generation. The mating chromosomes are selected based on some specific criteria.

3.3.2 Fitness Function

The productivity of any individual depends on the fitness value. It is the measure of the superiority of an individual in the population. The fitness value shows the performance of an individual in the population. Therefore, the individuals survive or die out according to the fitness or function value. Hence, the fitness function is the motivating factor in the GA.

3.3.3 Selection

The selection mechanism is used to select a middle solution for the next generation based on Darwin's law of survival. This operation is the guiding channel for the GA based on the performance.

There are various selection strategies to select the best chromosomes, such as roulette wheel, Boltzmann strategy, tournament selection, and selection based on rank.

3.3.4 Crossover

Crossover operation can be achieved by selecting two parent individuals and then creating a new individual tree by alternating and reforming the parts of those parents. The hybridization operation is a guiding process in the GA, and it boosts the searching mechanism.

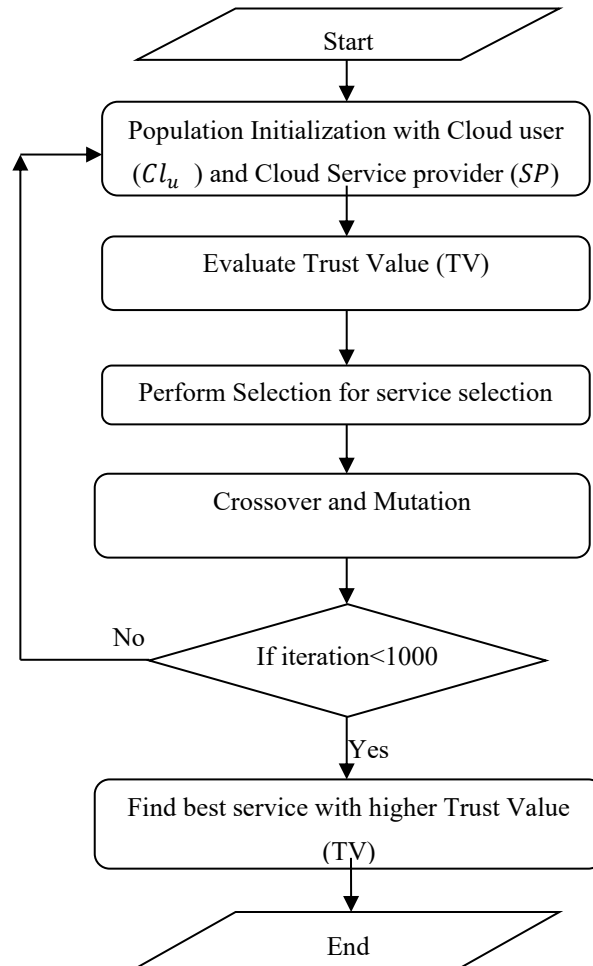


Figure 3: GA flowchart

3.3.5 Mutation

After crossover, mutation takes place. It is the operator that introduces genetic diversity in the population. The mutation takes place whenever the population tends to become homogeneous due to repeated use of reproduction and crossover operators. It occurs during evolution according to a user-defined mutation probability, usually set to a fairly low value. A mutation alters one or more gene values in the chromosome from its initial state. This can produce entirely new gene values being added to the gene pool. With these new gene values, the genetic algorithm may be able to produce a better solution than previously. Traditional GAs usually works well for single optimum problems, but they fail if they have to find multiple solutions. So, to overcome this problem, here Chaos GA is used.

3.4 Chaos Theory

Chaos is a bounded dynamic behavior that occurs in a deterministic nonlinear system. There are three main properties of the chaotic map, i.e., Ergodicity, Randomness, and Sensitivity to initial condition. The ergodicity property of chaos can ensure chaotic variables to traverse all states non-repeatedly within a certain range according to its own laws. So, this can be used as an optimization mechanism that avoids falling into a local minimum solution. The sensitivity to the initial state, one of the most important characteristics of chaotic systems, can ensure that there are not two identical new populations even if the two best-fit solutions obtained by sequential evolving procedures are very close. So, such a population not only reserves the best-fit chromosome, but also maintains population diversity. By using these properties, an effective approach was proposed for maintaining the population diversity and avoiding the search being trapped in a local optimum. In this paper, we use logistic and tent maps to generate the chaotic sequence. Chaotic sequences have been proven easy and fast to generate and store; there is no need for storage of long sequences. In addition, an enormous number of different sequences can be generated simply by changing its initial condition. Moreover, these sequences are deterministic and reproducible [15].

Logistic Map

Is one of the simplest forms of chaotic mappings. Basically, this map is a polynomial mapping of degree 2, whose equation is the following:

$$X_{n+1} = rX_n(1-X_n) \quad X_n \in (0,1) \quad (11)$$

Where r is a control parameter which is between 0 and 4.

Tent Map

Is one of the known chaotic systems, and it is a discrete-time dynamical system, whose equation is the following:

$$X_{n+1} = \begin{cases} \mu x_n & x_n < \frac{1}{2} \\ \mu(1 - x_n) & x_n \geq \frac{1}{2} \end{cases} \quad (12)$$

Where μ is a control parameter when $1 < \mu < 2$, the system exhibits chaotic behavior.

3.4.1 Chaos Genetic Algorithm

Recently, chaos theory and the generation of chaotic sequences instead of random ones have been adopted, which has led to very interesting results. GA has aroused intense interest, due to the flexibility, versatility, and robustness in solving optimization problems, which conventional optimization methods find difficult. One of the major disadvantages of the GA is its premature convergence and population initialization, especially when optimization problems have more local optima. To solve the initialization population in GA, chaos theory is introduced. The overall representation of CGA is shown in Figure 4. In this work, CGA is proposed to combine the concept of chaos with GA. By using these properties of chaos, propose a CGA based on two kinds of chaotic mapping. In improving the algorithm, use chaotic mapping instead of a random process. The standard GA uses random sequences in the initial population, crossover, and mutation. In the GA method, the initial population generated by a random approach might be unevenly distributed and away from the optimal solution. Hence, the algorithmic efficiency can be very low, and a larger number of iterations is needed to find the global optimum. Therefore, use the uniform distribution of the tent map to generate the initial population. Then, use the logistic map output instead of crossover and mutation each time a random number is needed. The flowchart of the proposed method is shown in Figure 4. This proposed methodology for service selection is compared with various existing methods, will be detailed discussion on the results and discussion section.

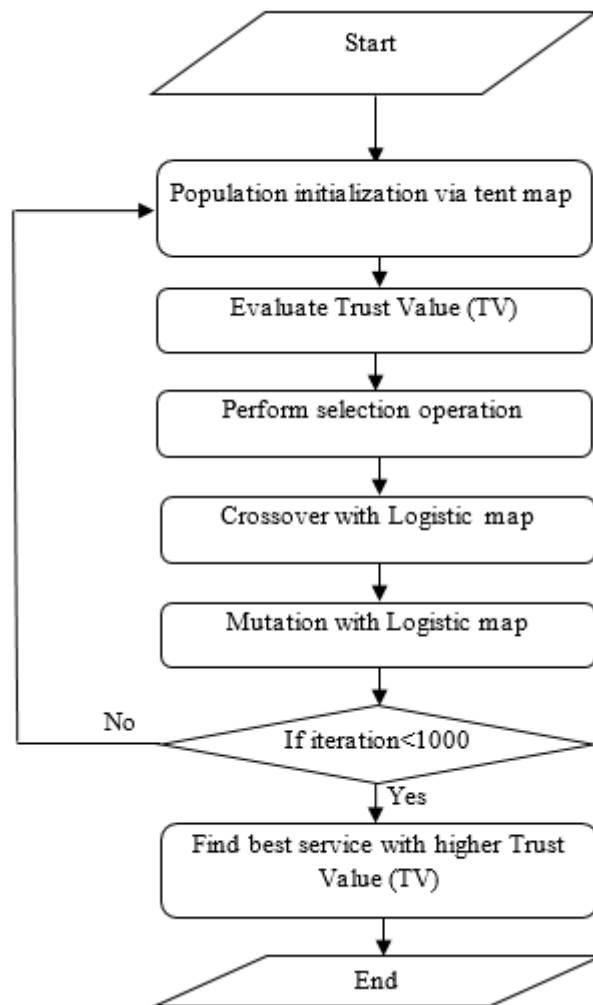


Figure 4: CGA flowchart

Input: Number of Service providers and number of users with chromosome: availability, reliability, data integrity, Turnaround Efficiency, Direct Trust, Recommended Trust

Output: Trust-based service selection

CGCA algorithm

Step 1: Start

Step 2: Population Initialization $P(i)$ with Cloud user (Cl_u) and Cloud Service provider (SP)

Step 3: If ($i=0$ to N)

3.1. Evaluate trust value (TV) for each population $P(i)$

3.2. Perform selection operation

3.3. Logistic map with crossover

3.4. Logistic map with mutation

Step 4: If (iteration < max iterations)

Step 5: Select the best cloud service with higher trust

Step 6: else go to step 2

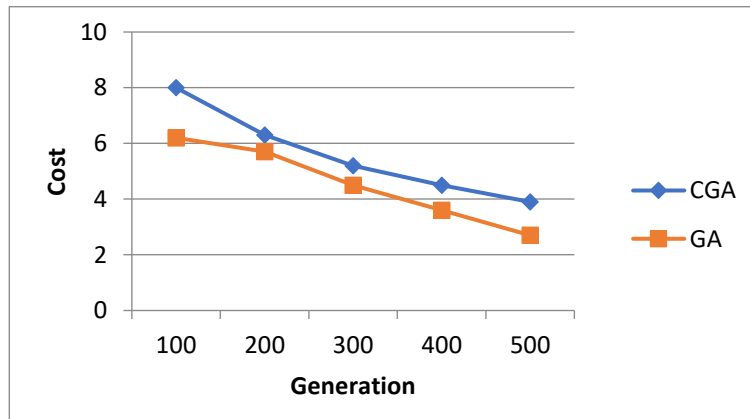
Step 7: end

4 Results and Discussion

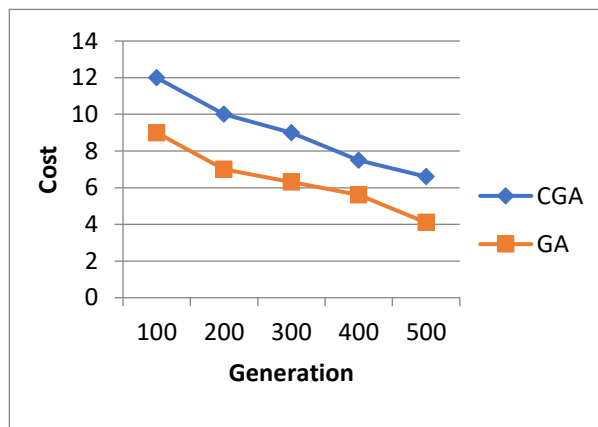
In this section, we are going to compare various existing techniques with the proposed TCSS-CGA for service selection in cloudlets in terms of trust value, response time, and throughput with respect to the number of services. The platform of the simulation environment is CloudSim toolkit, which is a simulation platform based on Java, that supports modeling and simulation of large-scale cloud computing data centers. Therefore, it is feasible to simulate the proposed model of cloud computing environments by CloudSim. Each service provider possesses a set of cloud components, and the set sizes of all service providers are uniformly distributed (i.e., from 1 to 10). One or more components can be combined to produce a cloud service. Use a simple case study example in this section. For this aim, three real cloud services, i.e., Amazon EC2, IBM, and GoGrid, are selected. Generate 100 Service Providers and 10000 Cloud Users, and 50 Kinds Of Different Cloud Components. The proposed work is tested on some famous benchmark functions, which Table 1 shows the main properties of the benchmark functions used in the experiments.

Table 1: Benchmark for simulation

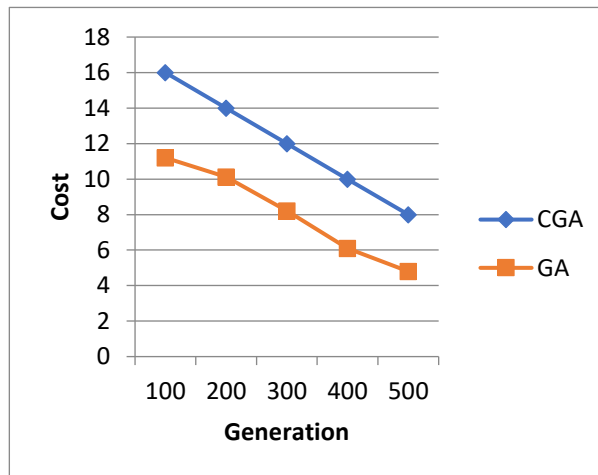
Function	Mathematical function	Range
Zakharov	$f(x) = \sum_{i=1}^n x_i^2 + \left(\sum_{i=1}^n 0.5ix_i \right)^2 + \left(\sum_{i=1}^n 0.5ix_i \right)^4$	$(-10,10)$
Ackley	$f(x) = 20 + e - 20 \exp \left(-0.2 \left(\sqrt{\frac{1}{n} \left(\sum_{i=1}^n x_i^2 \right)} - \exp \left(\frac{1}{n} \left(\sum_{i=1}^n \cos 2\pi x_i \right) \right) \right) \right)$	$(-15,15)$
Rosenbrock	$f(x) = \sum_{i=1}^{n-1} (100(x_i^2 - x_{i+1})^2 + (x_i - 1)^2)$	$(-20,20)$
Rastrigin	$f(x) = \sum_{i=1}^{n-1} (x_i^2 - 10 \cos 2\pi x_i + 10)$	$(-4.15,4.15)$



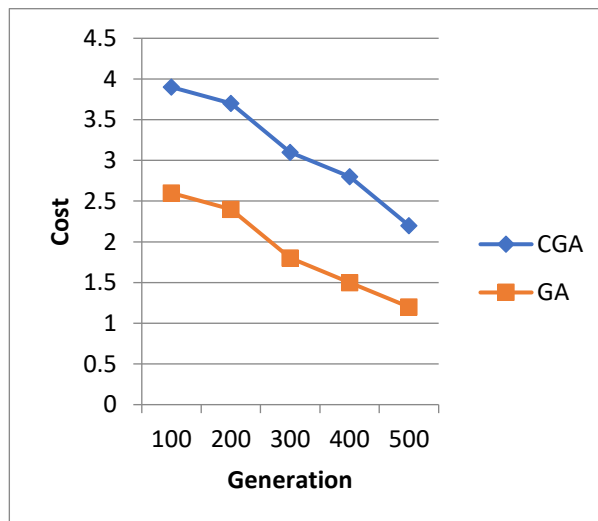
(a) Fitness value for CGA and GA on the Zakharov benchmark function



(b) Fitness value for CGA and GA on Ackley benchmark function



(c) Fitness value for CGA and GA on the Rosenbrock benchmark function



(d) Fitness value for CGA and GA on the Rastrigin benchmark function

Figure 5 (a,b,c,d): Fitness value for CGA and GA on 4 benchmark functions

Figure 5 shows the performance of the CGA and GA for solving four functions. Obviously, the convergence speed of CGA is faster than that of basic GA.

The existing methods are Trust-Based Cloud Service Provider Selection (TBCSPS) and Trust Evaluation Selecting Service Provider (TESSP). Figures 6, 7, and 8 show the comparison of trust accuracy, response time, and throughput with respect to the number of services.

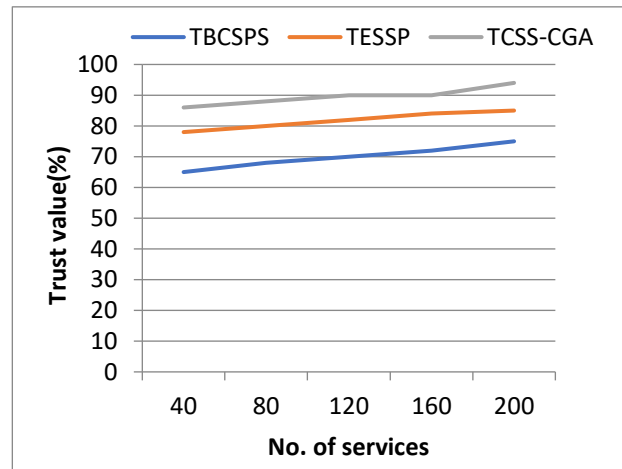


Figure 6: Comparison of Trust Accuracy vs. Number of Services

From Figure 6, the proposed TCSS-CGA has a higher trust value when compared to the other two existing methods, as shown. The proposed TCSS-CGA has 90% at 160 of number of services as a trust value, whereas TBCSPS and TESSP have 72% and 84% as trust values at service number 160, respectively.

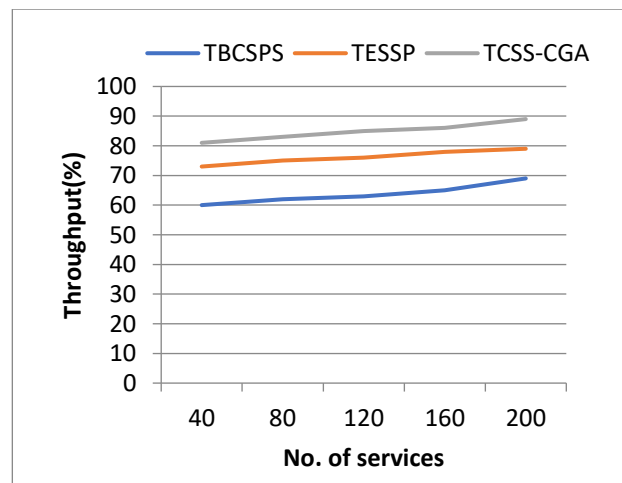


Figure 7: Comparison of throughput vs. Number of services

From Figure 7, the proposed TCSS-CGA has a higher throughput when compared to the other two existing methods, as shown. The proposed TCSS-CGA has 86% as throughput at service number 160, whereas TBCSPS and TESSP have 65% and 78% as throughput at service 160, respectively.

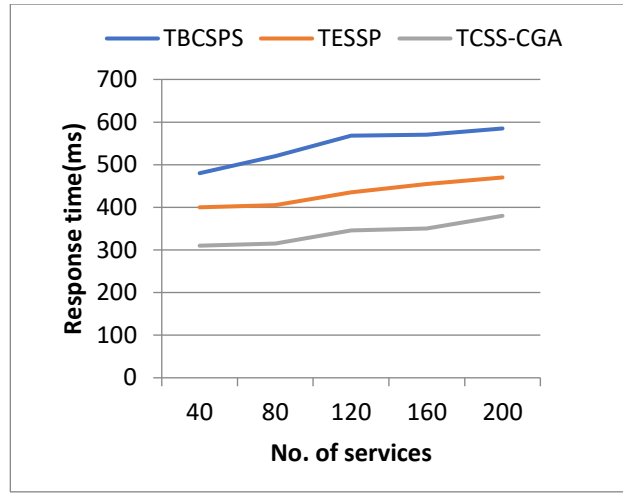


Figure 8: Comparison of response Time vs. Number of services

From Figure 8, the proposed TCSS-CGA has a lower response when compared to the other two existing methods, as shown. The proposed TCSS-CGA has 350 ms at a number of services 160, whereas TBCSPS and TESSP have 570 ms and 455 ms as response time at service number 160, respectively.

Assess the effectiveness of cloudlet service selection methods with Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which are defined as:

$$MAE = \frac{\sum_{j,i} |Q_{j,i} - Q_{j,i}^{pre}|}{N} \quad (13)$$

$$RMSE = \sqrt{\frac{\sum_{j,i} (Q_{j,i} - Q_{j,i}^{pre})^2}{N}} \quad (14)$$

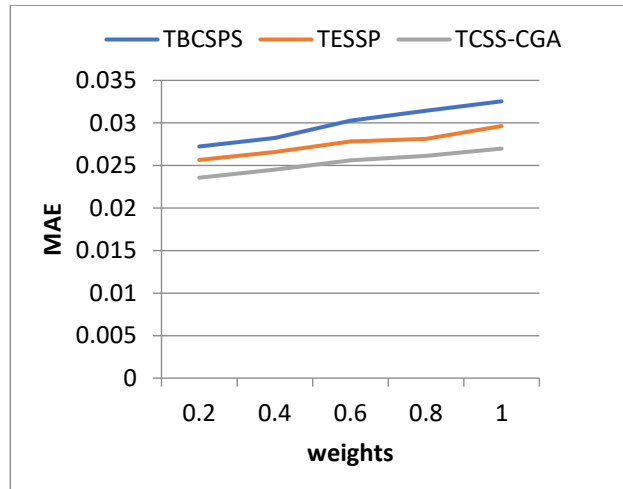


Figure 9: MAE comparison with respect to weights

Figure 9 shows the MAE comparison results of the proposed TCSS-CGA approach has a lower MAE when compared to the other two existing methods. The proposed TCSS-CGA approach provides

a lower MAE of 0.02698 at a weight value of 1.0, whereas TBCSPS and TESSP have 0.03254 and 0.02963, respectively.

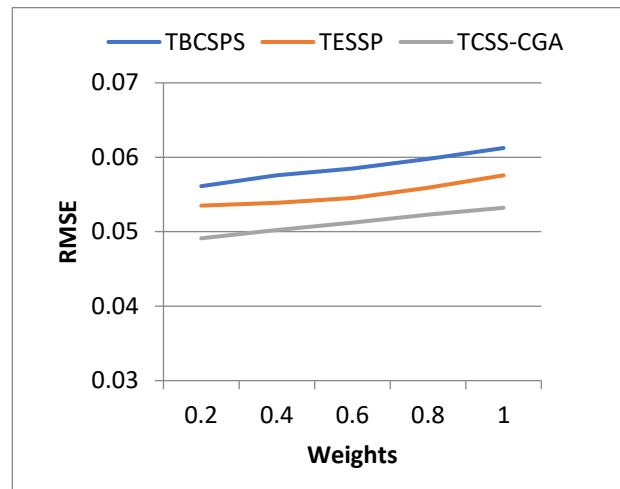


Figure 10: RMSE comparison with respect to weights

Figure 10 shows that the RMSE comparison results of the proposed TCSS-CGA approach have a lower RMSE when compared to the other two existing methods. The proposed TCSS-CGA approach provides a lower RMSE of 0.05322 at a weight value of 1.0, whereas TBCSPS and TESSP have 0.06125 and 0.05758, respectively.

5 Conclusion and Future Work

In this study, a trust-based cloudlet selection would take place for a cloud service provider. Various trust evaluations are done by the trust management system, like availability, reliability, data integrity, and turnaround efficiency, and performance analysis, like time response and power consumption, for the selection of service in a cloudlet. Here, the fitness values are optimized by using CGA to improve the performance of the system. Experimentally proven that the proposed TCSS-CGA algorithm would give a better performance for a selection of services when compared to all other existing methods. In the future, trust evaluation will include offloading.

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